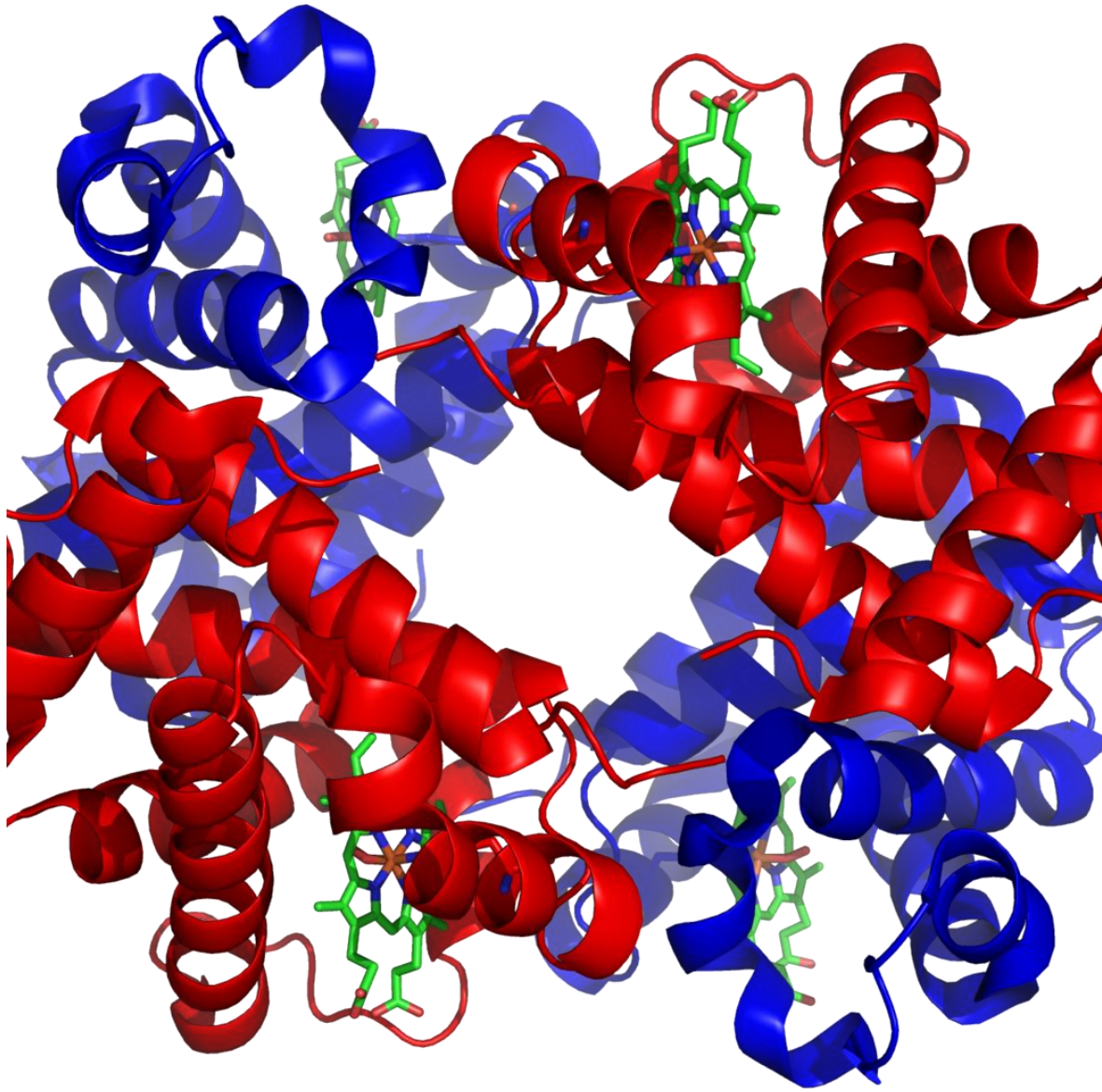
DP Biology – B1.2 (HL extension)

Protein structure

- How does the order of amino acids produce such a variety of protein forms and functions?
- How do chemical and physical conditions affect protein molecules?



HL statements B1.2.6–12. Textbook pp. 210–222.
Image: haemoglobin, four chains with haem groups (green). Richard Wheeler, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	B1.2.6–8	Explain how R-groups give proteins their properties, and how primary structure leads to secondary structure
2	B1.2.9–10	Explain the bonds that hold tertiary structure, and where hydrophobic and hydrophilic amino acids end up
3	B1.2.11–12	Compare quaternary structure in insulin, collagen and haemoglobin, and relate globular and fibrous forms to function

The big picture: four levels of structure



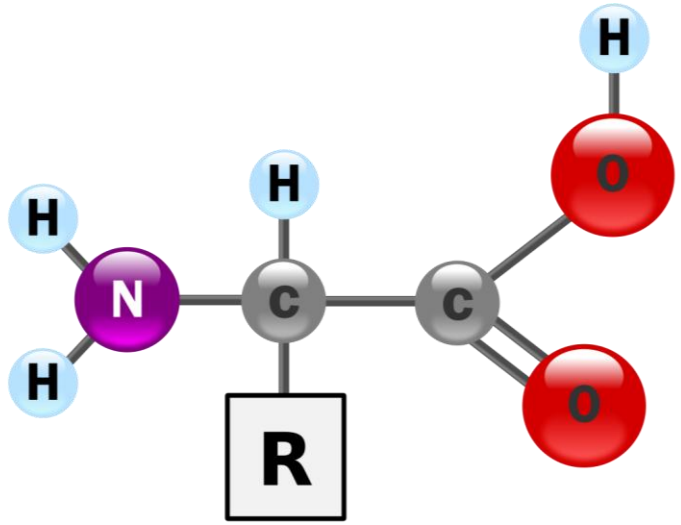
The **sequence** of R-groups decides how a chain folds. Each level is held by particular **bonds**, and heat or pH can break them.

Big question

How can 20 amino acids make both a soft hormone and the toughest tissue in your body?

R-groups make the difference

B1.2.6



General structure of an amino acid. Public domain.

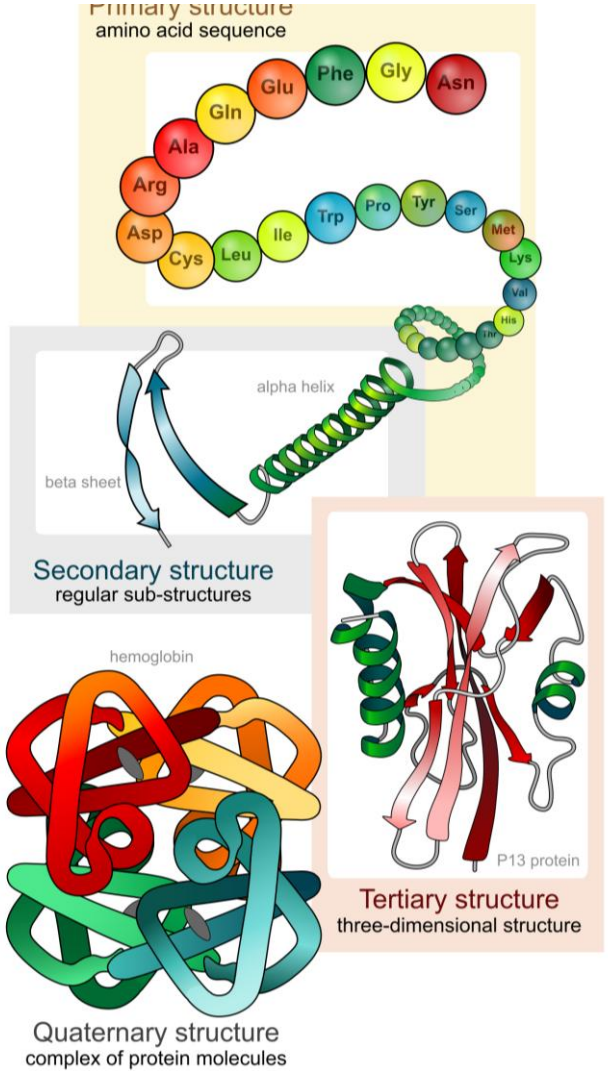
R-group	Properties
Non-polar	Hydrophobic
Polar	Hydrophilic: hydrogen bonds
Acidic	-COOH loses H^+: negative
Basic	Amine gains H^+: positive

All amino acids share a backbone; only the **R-group** differs in size, charge and polarity.

Key idea
20 R-groups in any order give almost unlimited sequences.

Primary structure decides the shape

B1.2.7



Levels of protein structure. LadyofHats, public domain.

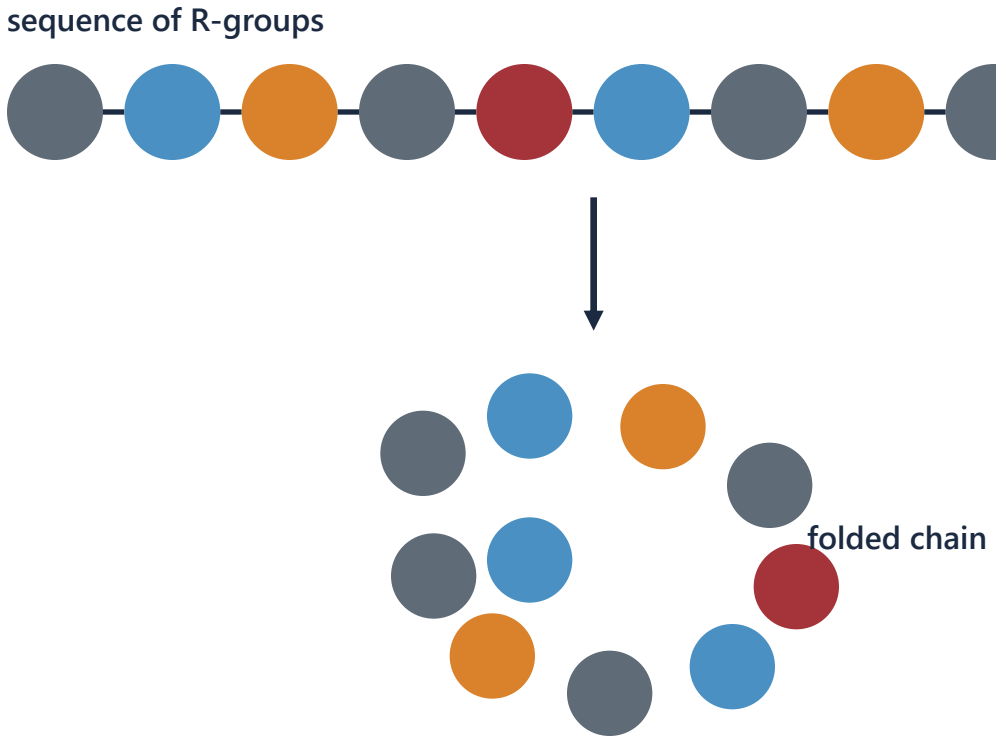
The **primary structure** is the amino acid sequence, set by the gene.

It fixes where each R-group sits, so it decides which can interact and how the chain **folds**.

Think

Why can a computer predict shape from the sequence, but not from a count of amino acids?

From sequence to fold



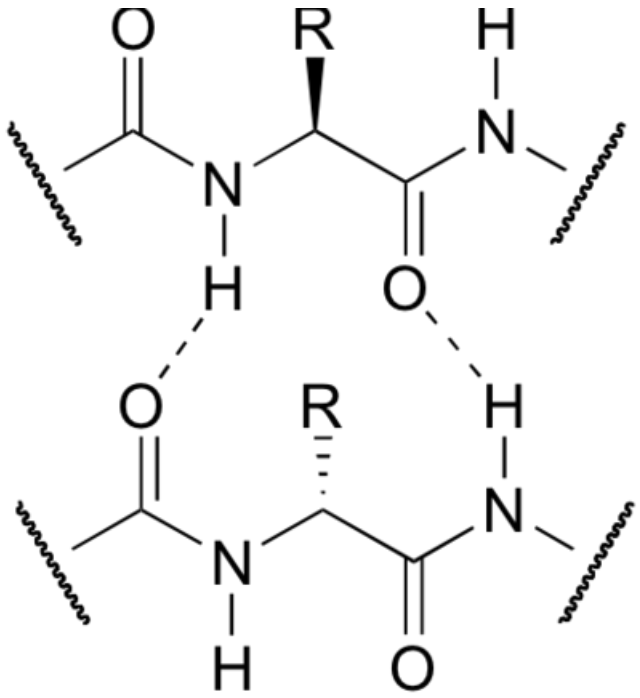
	R-group
Grey	Non-polar: inside
Blue	Polar: outside
Orange	Charged +
Red	Charged -

Key idea

Change one amino acid and the fold may change: sickle-cell haemoglobin.

Secondary structure: helices and pleated sheets

B1.2.8



hydrogen bonding in a β-strand

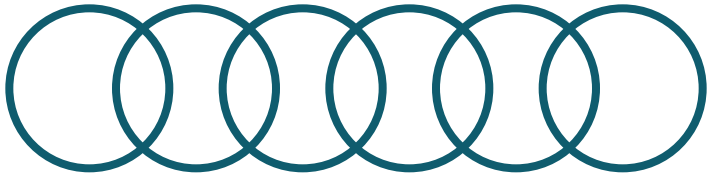
Hydrogen bonds between C=O and N-H in the backbone. CC BY-SA

	Alpha helix	Beta-pleated sheet
Shape	Coil	Strands side by side
H bonds	C=O to N-H four along, same chain	C=O to N-H of neighbouring strands

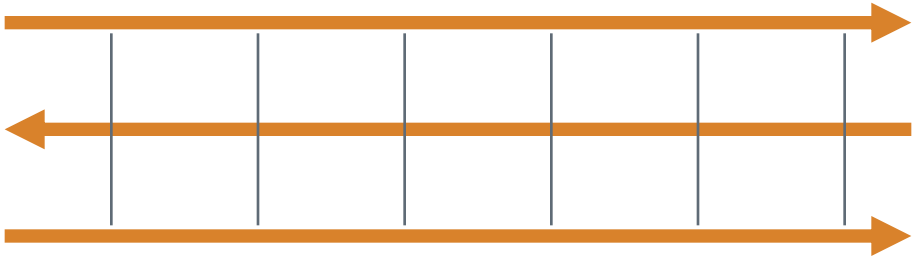
Common mistake

Secondary structure uses backbone hydrogen bonds, not R-groups.

Drawing helices and sheets



alpha helix (side view)



beta-pleated sheet: grey = hydrogen bonds

Exam tip

Helix: coil drawn as loops or a ribbon. Sheet: arrows side by side, pointing alternate ways, with hydrogen bonds between them.

Practice: R-groups and structure

B1.2.6–8

Classify

An R-group with -CH2-CH3. And with -COOH?

Explain

Why sequence decides shape.

Distinguish

Bonds in secondary vs tertiary structure.

Use

R-group, polar, backbone, hydrogen bond

Check-in: class 1

B1.2.6–8

1. Classify an R-group with an extra -COOH.
2. Explain why primary structure determines shape.
3. State the atoms joined by hydrogen bonds in an alpha helix.
4. Distinguish an alpha helix from a beta-pleated sheet.
5. Explain why secondary structures are stable.

How to use this

Answer without your notes, then check against slides 3–9.

Tertiary structure: bonds between R-groups

B1.2.9

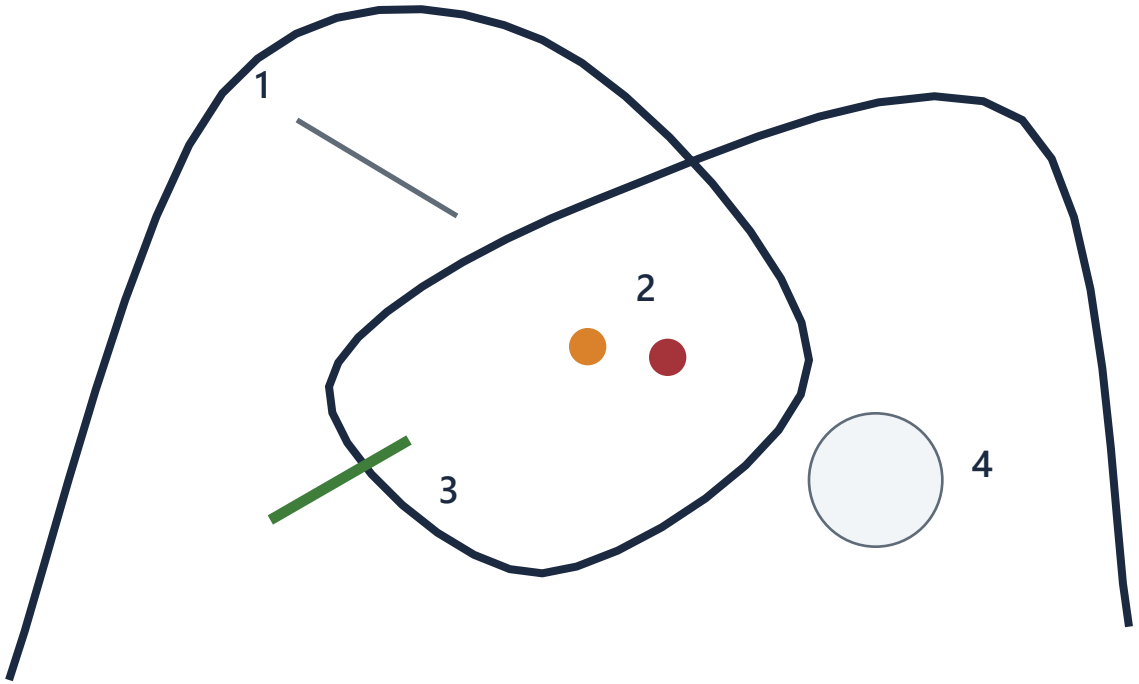
Bond	Between	Strength and sensitivity
Hydrogen bonds	Polar R-groups	Weak; broken by heat
Ionic bonds	+ amine and − carboxyl R-groups	Moderate; changed by pH
Disulfide bonds	-SH groups of two cysteines	Strong covalent
Hydrophobic interactions	Non-polar R-groups clustered away from water	Weak alone; many together

Charge depends on pH

Amines gain H^+ and carboxyls lose H^+, so pH changes ionic bonds and can denature proteins.

Where the bonds are

B1.2.9



1 hydrogen bond (grey)

2 ionic bond (+/-)

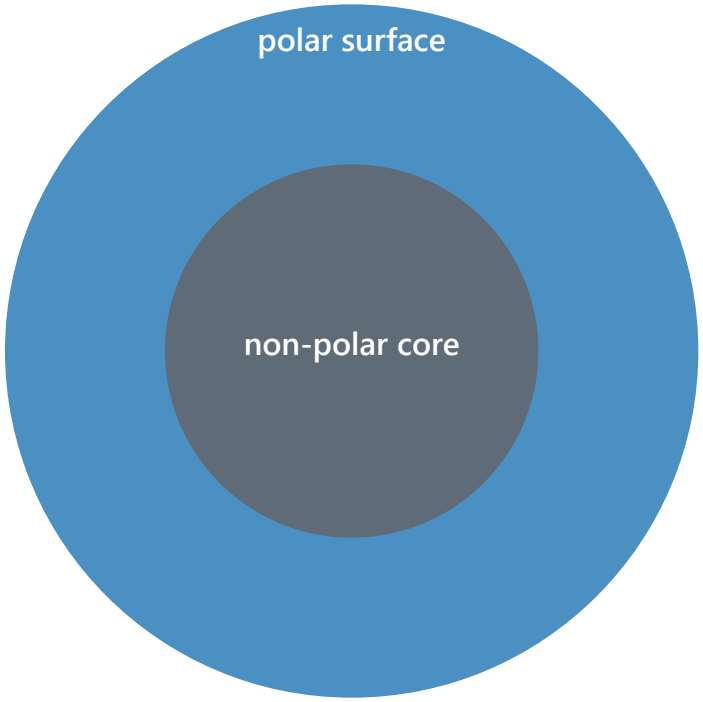
3 disulfide bond (green)

4 hydrophobic cluster

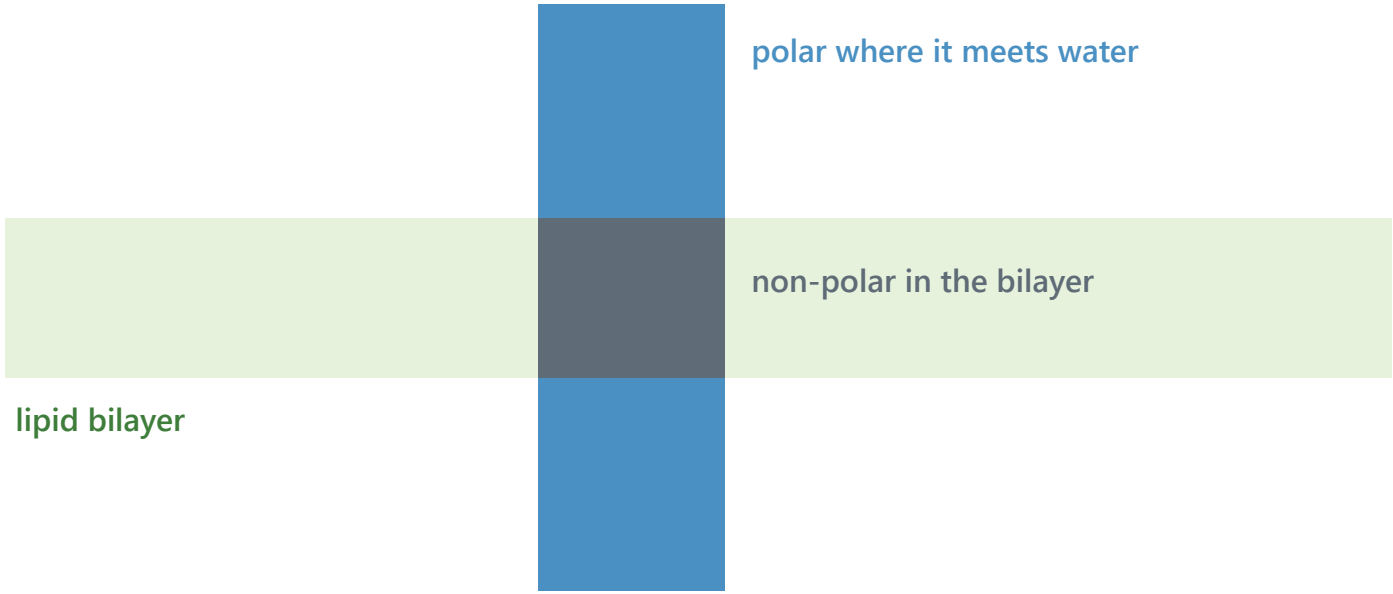
Heat breaks weak bonds; pH changes ionic bonds.

Hydrophobic inside, hydrophilic outside

B1.2.10



soluble globular protein



Key idea

Non-polar R-groups avoid water; polar ones face it. Channels are lined with polar R-groups.

Predict: a run of non-polar amino acids

B1.2.10

Activity

A protein has a stretch of about 20 non-polar amino acids. Predict where in the cell this stretch is, and explain. What if one became polar?

Hint

About 20 amino acids in an alpha helix is roughly the thickness of a membrane.

Working with data: heat and a protein’s shape

B1.2.9

Enzyme activity after 10 min at each temperature, measured at 25 °C. Invented values; realistic pattern.

Pre-heated / °C	Activity left / %
25	100
40	98
50	80
60	35
70	5
80	0

1. Describe the effect of heating.
2. Explain it using tertiary bonds.
3. Why is activity lost despite disulfide bonds?
4. Suggest how a hot-spring enzyme might differ.

Practice: bonds and folding

B1.2.9–10

Explain

Why boiling denatures insulin but leaves S-S bonds.

Predict

Effect of pH 2 on ionic bonds.

Locate

Non-polar R-groups in a channel protein.

Task

Open insulin on PDB-101, colour by hydrophobicity, and find a disulfide bond.

Check-in: class 2

B1.2.9–10

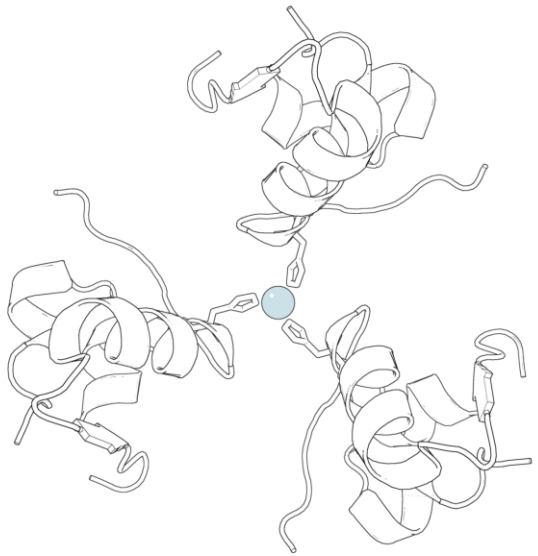
1. List four bonds in tertiary structure.
2. State which is covalent.
3. Explain how pH can denature a protein.
4. State where non-polar amino acids are in a soluble protein.
5. Describe the parts of an integral protein in the bilayer.

How to use this

Answer without your notes, then check against slides 11–16.

Quaternary structure

B1.2.11



Insulin hexamer. Public domain.

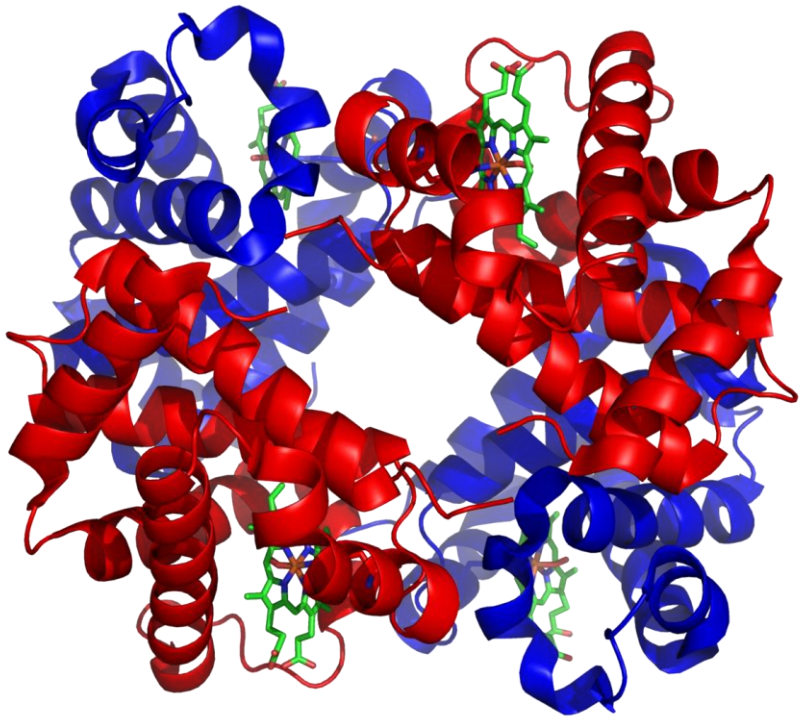
Protein	Polypeptides	Conjugated?
Insulin	Two chains joined by S-S bonds	No
Collagen	Three chains: triple helix	No
Haemoglobin	Four chains, each with haem	Yes: prosthetic group

Definitions

Quaternary: two or more polypeptides joined. Conjugated: has a non-polypeptide prosthetic group.

Haemoglobin: a conjugated protein

B1.2.11



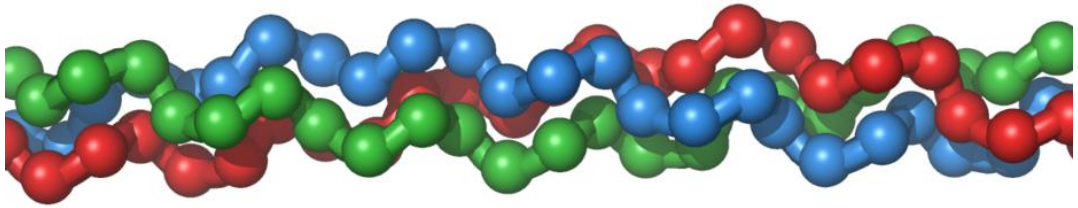
Haemoglobin: alpha red, beta blue, haem green. CC BY-SA 3.0.

Four chains: two alpha, two beta. Each holds a **haem** with an **iron ion** that binds one O_2.

When one O_2 binds, the shape shifts and the next binds more easily: **cooperative binding**.

Collagen: a fibrous protein

B1.2.12



Collagen triple helix. CC BY-SA 3.0.

Link
Vitamin C is needed for collagen: scurvy.

Collagen gives strength to skin, tendons and bone.

Every third amino acid is glycine, the smallest, so three chains wind tightly into a **triple helix**; cross-linked fibres resist pulling.

Form and function: globular and fibrous

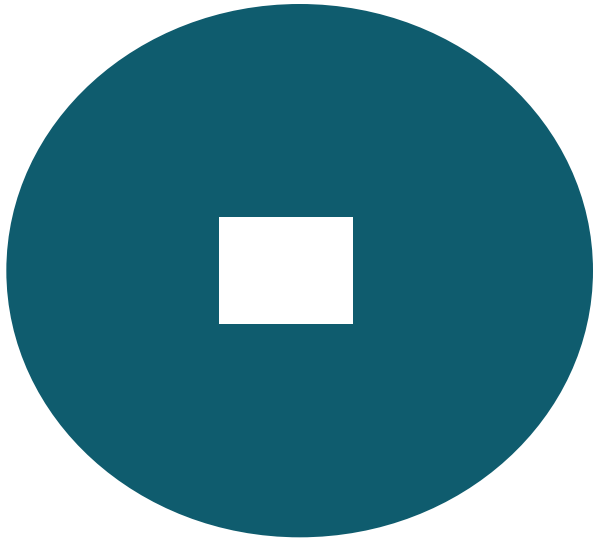
B1.2.12

	Insulin (globular)	Collagen (fibrous)
Shape	Compact, roughly spherical	Long triple helix
Solubility	Soluble in plasma	Insoluble
Sequence	Varied; precise fold	Repetitive; glycine every third
Function	Hormone: binds a receptor	Strength in tissues

Big question

Same building blocks; sequence makes one a precise key and the other a rope.

Picture the difference



globular: binding site



fibrous: three chains wound together

Try it

Explain why a fibrous protein would be no use as a hormone.

Practice: quaternary and form

B1.2.11–12

Define

Quaternary structure; conjugated protein.

Explain

Why glycine is every third amino acid in collagen.

Compare

Insulin and collagen: three differences.

Use

polypeptide, prosthetic group, triple helix, soluble

Check-in: class 3

B1.2.11–12

1. Define quaternary structure.
2. Explain why haemoglobin is conjugated.
3. Describe cooperative binding.
4. Explain glycine's role in collagen.
5. Distinguish globular and fibrous proteins.

How to use this

Answer without your notes, then check against slides 18–23.

Common mistakes to avoid

Not this	But this
Secondary structure uses R-group bonds.	Backbone C=O and N-H hydrogen bonds.
Disulfide bonds are hydrogen bonds.	They are strong covalent bonds.
All proteins have quaternary structure.	Only those with 2+ polypeptides.
Insulin is conjugated.	Haemoglobin is; insulin has no prosthetic group.
Denaturation breaks peptide bonds.	It breaks the bonds holding the 3D shape.

Summary: protein structure

B1.2.6–12

	Key idea
B1.2.6	R-groups: non-polar, polar, acidic, basic
B1.2.7	Sequence decides fold
B1.2.8	Helix and sheet: backbone H bonds
B1.2.9	Tertiary: H, ionic, S-S, hydrophobic
B1.2.10	Non-polar inside or in membranes
B1.2.11	Quaternary; conjugated haemoglobin
B1.2.12	Globular insulin vs fibrous collagen

Check that you can...

1. **Classify** R-groups
2. **Explain** how primary structure decides shape
3. **Describe** helices and sheets
4. **Outline** tertiary bonds and effects of heat and pH
5. **Compare** insulin, collagen and haemoglobin
6. **Relate** globular and fibrous forms to function

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Name the type of bond that stabilizes an alpha helix, and the atoms it joins.
2. State where non-polar amino acids are found in a soluble globular protein.
3. Explain why haemoglobin is called a conjugated protein.

Image credits

1GZX Haemoglobin.png – Zephyris at English Wikipedia – CC BY-SA 3.0

AminoAcidball.svg – GYassineMrabet – Public domain

Main protein structure levels en.svg – LadyofHats – Public domain

Both helix and the beta-sheet structures are held by specific hydrogen-bonding interactions.png – Unknown author – CC BY-SA 4.0

Insulin hexamer 4INS.png – Fvasconcellos (talk · contribs) – Public domain

Collagentriplehelix.png – Author not stated – CC BY-SA 3.0

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses